

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads to the
Interstate Transmission System

Docket No. RM26-4-000

JOINT COMMENTS OF THE DISTRIBUTED CAPACITY PARTIES

In response to the above captioned Advanced Notice of Proposed Rulemaking (“Large Load Interconnection ANOPR”), Cloverleaf Infrastructure (“Cloverleaf”), Spark Community Investment Company d/b/a Sparkfund (“Sparkfund”), and Voltus, Inc., (“Voltus”)¹ (collectively, the “Distributed Capacity Parties”) submit the following comments highlighting the capabilities of demand-side and advanced technologies to support the rapid interconnection of large loads. In light of this *historic* moment, the Distributed Capacity Parties respectfully urge the Federal Energy Regulatory Commission (“FERC” or “Commission”) in any subsequent rulemaking² to direct the Regional Transmission Organizations/Independent System Operators (“RTO/ISOs”) to develop mechanisms for granting expedited interconnection studies to large loads that provide

¹ Voltus is a leading distributed energy resource (“DER”) aggregator that provides a platform to enable consumers to connect their distributed energy resources including demand response to energy markets. Voltus identifies distributed energy resources, develops the aggregations, bids into wholesale markets, dispatches DERs, and performs the settling and reporting. Voltus operates aggregations of DERs on behalf of consumers in every ISO/RTO market in the United States.

² The Distributed Capacity Parties acknowledge that parties may contest FERC’s jurisdiction to issue a Notice of Proposed Rulemaking (“NOPR”) on large load interconnection. The Distributed Capacity Parties do not take a position on FERC’s jurisdiction.

curtailable and/or dispatchable distributed assets (aka Distributed Capacity³) capable of mitigating potential system constraints attributable to the new large loads. The ANOPR Seventh Principle⁴ should include incremental Distributed Capacity in the form of load flexibility, storage, and/or generation capacity whose location can address capacity and transmission constraints projected to be caused by the new large loads, even if such capacity is located on the distribution system.

Distributed capacity technologies include dispatchable resources such as load flexibility, storage (e.g., 1-5 MW distribution-connected batteries), and distributed generation, as well as advanced technologies like grid enhancing technologies. Distributed capacity technologies may be located on the distribution system,⁵ whether behind the meter, in front of the meter, or “around” the meter. Distributed capacity assets currently provide critical grid services at wholesale,⁶ making them a critical tool *today* to accelerate large load interconnections.

I. INTRODUCTION

Recognizing the impacts of large load additions on the transmission system and wholesale rates, and responding to the observable step-change in the volumes and types of

³ See, e.g., Andy Lubershane, “The price of power grids” (Apr. 7, 2025), <https://steelforfuel.substack.com/p/the-price-of-power-grids>.

⁴ The Seventh Principle in the Large Load Interconnection ANOPR is: “Seventh, the interconnection study of large loads that agree to be curtailable and hybrid facilities that agree to be curtailable and dispatchable should be expedited. The system operator’s ability to control such facilities through curtailment and/or dispatch must be sufficient for the system operator to integrate the facility into both operations and system planning.”).

⁵ FERC has defined a DER as “any resource located on the distribution system, any subsystem thereof or behind a customer meter. These resources may include, but are not limited to, electric storage resources, distributed generation, demand response, energy efficiency, thermal storage, and electric vehicles and their supply equipment.” Order No. 2222, 172 FERC 61,247, P 1 n.1 (2020). In the same order, FERC further noted, “The revised definition of distributed energy resource that we adopt in this final rule is technology-neutral, thereby ensuring that any resource that is technically capable of providing wholesale services through aggregation is eligible to do so, which enhances competition in the RTO/ISO markets and, in turn, helps to ensure that these markets produce just and reasonable rates.” *Id.* at P 114.

⁶ See, e.g., Stephanie Bieler, Cara Goldenberg, Avery McEvoy, Katerina Stephan, Alex Walmsley, *Aggregated Distributed Energy Resources in 2024: The Fundamentals*, NARUC/NASEO/RMI p. 30 (July 2024), available at <https://pubs.naruc.org/pub/98FBE453-02C0-1FE3-0249-3A456BA1E3E7> (indicating a minimum set of bulk power system grid services that aggregated distributed energy resources provide today).

planned new loads, the instant ANOPR seeks to standardize the large load interconnection process to ensure “timely and orderly” interconnection. Several of the fourteen outlined ANOPR principles deal with the study of hybrid or co-located load and generation arrangements. But traditional generators will take years to build, even with massive capital investments by hyperscalers and others seeking to connect new large loads.⁷

Distributed capacity assets are installed *today* across critical footprints, are readily installable en masse, and may be aggregated into utility-scale assets on a timeline of months rather than years.⁸ Wholesale market demand response—which is primarily commercial and industrial load—has been contributing to affordability and reliability for over a decade and can scale rapidly. In 2023 alone, the Southwest Power Pool (“SPP”) added approximately 431 MW new demand response resource capability, equivalent to a large power plant.⁹ Another 500 MW power plant could be added by interconnecting to the distribution system a mere 200 batteries of 1-5 MW size. Residential resources can also be mobilized: a recent report by Rewiring America identifies an additional 93 GW of needed data center capacity through 2029¹⁰ but also finds that “hyperscalers could more than meet their total planned capacity needs by paying for battery storage as well as rooftop solar for homes well suited for it.”¹¹ Proving that distributed residential

⁷ See, e.g., Tom Rutigliano and Claire Lang-Ree, *Large Load CIFP: NRDC Solution Components*, Natural Resources Defense Council p. 2 (Oct. 14, 2025), <https://www.pjm.com/-/media/DotCom/committees-groups/cifp-lla/2025/20251014/20251014-item-03c---nrdc-proposed-options.pdf> (presentation given, as part of PJM Interconnection LLC’s recent Large Load Addition Critical Issue Fast Path process, highlighting time-lags for new generation projects and associated price relief, and noting that “a new project that was ready to go the day after the 2025/26 BRA results will not realistically lower capacity prices until 2033/34.”).

⁸ See Cora Wyent, Miki Verma, and Wael Kanj, *Homegrown energy: How household upgrades can meet 100 percent of data center demand growth*, Rewiring America (2025), available at <https://www.rewiringamerica.org/research/homegrown-energy-report-ai-data-center-demand>; see also Jesse Cohen, Lauren Shwisberg, Mark Dyson, *How Virtual Power Plants Can Help the United States Win the AI Race* (Nov. 6, 2025), available at <https://rmi.org/how-virtual-power-plants-can-help-the-united-states-win-the-ai-race>.

⁹ FERC, 2024 Assessment of Demand Response and Net Metering, at 14 (Dec. 2024), available at <https://www.ferc.gov/news-events/news/ferc-staff-issues-2024-assessment-demand-response-and-advanced-metering>.

¹⁰ Wyent et al. at 2.

¹¹ Wyent et al. at 5.

assets can add capacity fast, Arizona Public Service’s Cool Rewards thermostat program has brought 40 MW of capacity annually in recent years, and Ontario tapped 90 MW of household flexibility in just six months.¹² If the Commission truly wants to bring all American resources to bear in service of interconnecting large loads, Distributed Capacity is one of the only resource types that can be enlisted on a timeline that matches the urgency of the challenge at a scale that counts.

To fully realize these benefits at wholesale and meet the intent of the ANOPR—and consistent with FERC’s non-discriminatory treatment of such resources as developed in Order Nos. 719, 745, 841, and 2222¹³—any subsequent rulemaking should promulgate guidelines on how distributed resources can be leveraged to reduce grid constraints in key regions, thereby freeing headroom for appropriately sited new large loads. The following section of these comments offers guidelines within the framework of the ANOPR’s fourteen principles.

II. GUIDELINES TO ENABLE EFFICIENT UTILIZATION OF DISTRIBUTED CAPACITY

A. Interconnection Reform Should Be Defined to Solve System Constraints at the Level at Which They Arise.

Large loads face interconnection delays for several reasons, including insufficient capacity to supply new load reliably, the need for costly network upgrades to bring incremental peak power to the location of new loads, and the additional time required for Transmission Owners (“TOs”) to study the foregoing system impacts. The current ANOPR targets these mechanisms under the basic premise that *load and generation configurations that add less strain*

¹² Cohen et al., *supra* note 8.

¹³ Order No. 719, 125 FERC 61,071 (2008); Order No. 745, 134 FERC 61,187 (2011); Order No. 841, 162 FERC 61,127 (2018); Order No. 2222, 172 FERC 61,247 (2020).

to the transmission system should be prioritized for interconnection. To achieve this prioritization, the ANOPR states that load and generation should be studied for interconnection holistically (ANOPR Principle Three), that interconnecting load bringing generation should be studied based on the actual volume of its net withdrawal or injection needs (ANOPR Principle Five), and that configurations imposing less system stress should be incentivized with the “carrot” of expedited interconnection (ANOPR Principle Seven).

These comments endorse the above principles while expanding on them to ensure that “electrically proximate” or “adjacent” grid assets—assets that can mitigate the constraints imposed by new loads *where those constraints actually occur*—are also considered in study prioritization, netted against load withdrawals for study purposes, and included in speed-to-power incentive structures.¹⁴

B. New Large Loads Should Be Studied Together With Distributed Capacity in the Relevant Footprint.

New large loads that bring their own capacity—in whatever form—should be studied together with that capacity. Data center developers, hungry for solutions, have already proposed Bring-Your-Own Capacity constructs.¹⁵ To this end, the existing ANOPR Principle Three, “load and hybrid facilities should be studied together with generating facilities,” should explicitly

¹⁴ The effective use of distributed capacity to support large-load interconnections depends on close coordination with the distribution utilities that own and operate the underlying systems. Because processes to integrate distributed capacity are fundamentally state-jurisdictional, utilities retain responsibility for ensuring that any distributed resource deployed to mitigate local or transmission constraints can be safely integrated into the distribution system. For this reason, FERC could direct the RTOs/ISOs to consider distributed capacity in connecting large loads, while specifics would likely involve contractual arrangements between a distribution utility, the large load, and the distributed capacity provider.

¹⁵ See Jeff St. John, *Can crowdsourcing help solve the data-center power crunch?* (last accessed Nov. 21, 2025), available at <https://www.canarymedia.com/articles/virtual-power-plants/voltus-supply-data-center-energy-needs-cloverleaf> (“The right way to serve data center load quickly, at scale, and less expensively and more sustainably, is to leverage the existing resources on the grid as efficiently as possible,” said Brian Janous, Cloverleaf’s chief commercial officer.”).

include study of *incremental resources in the relevant grid area*,¹⁶ whether or not those resources look like traditional utility-scale power plants. The ANOPR notes the rationale that “siting a large load near or at the same point of interconnection as a new generating facility could reduce the network upgrades needed to interconnect only the load or only the generating facility.” Similarly, siting a new large load while considering the location of installed but underutilized grid resources and proposed aggregations of the same can save on network upgrades. For instance, “latent” demand flexibility can meet incremental capacity needs by reducing power draw from the grid, thereby freeing up network capacity. Operationally, a hyperscaler could contract with a battery installer to develop new batteries in the area of a relevant transmission and/or capacity constraint. Studying this contracted project together with the load’s interconnection could be within the purview of ANOPR Principle Three.

C. Large Loads That Bring Locationally Relevant New Capacity Should Receive Expedited Interconnection.

The existing ANOPR Principle Seven asserts that “the interconnection study of large loads that agree to be curtailable and hybrid facilities that agree to be curtailable and dispatchable should be expedited.”¹⁷ This Principle recognizes the mitigated transmission *and capacity* impacts of sites whose withdrawal or injection to the grid is responsive to a signal from the system operator during transmission-constrained periods and system peaks, and finds that load that is willing today to interconnect under these curtailment/dispatchability conditions should be given expedited treatment.

However, to realize these benefits, new capacity supporting large loads need not be physically co-located at the facility, but can be *distributed across the geographic area within*

¹⁶ The “relevant footprint” would vary by RTO/ISO.

¹⁷ Large Load Interconnection ANOPR at P. 24.

which the new load will impact system constraints, provided that the system operator has the requisite level of control to integrate the facility into operations and system planning.¹⁸

ANOPR Principle Seven should be broadened to leverage all available resources. The Principle as written does not recognize the reduced impact of an interconnecting load that brings on (contracts for) demand-side flexibility or capacity not physically “within the fence-line” of the facility¹⁹ but still appropriately located to relieve constraints. Such capacity can be considered in tranches: a new load may have local impacts that require some amount of flexibility or capacity near or at the same point of interconnection, while it is sufficient to meet additional constraints with resource(s) within the load’s delivery zone or wholesale market region. Subject to these constraints, capacity—including batteries, generation, and demand response (DR)—should be geographically transferable in the form of capacity credits as defined by the wholesale market, or within the TOs system planning paradigm.

D. Existing Barriers to Distributed Capacity Must Be Addressed

To unlock the capabilities of existing distributed capacity, barriers to participation by aggregated resources in capacity, energy, and ancillary service markets should be rectified. Delays in Order No. 2222 implementation, utility data access issues, and other challenges significantly limit the development of latent flexibility today. As this ANOPR demonstrates, the need to unlock latent load flexibility and distributed generation that can rapidly alleviate transmission and capacity scarcity is more urgent than ever. In today’s unprecedented load growth environment, and with new utility scale generation and network upgrades often years

¹⁸ Such capacity configurations may be termed “electrically proximate” or “adjacent” and should be understood as an extension of co-location.

¹⁹ See, e.g., Elizabeth K. Whitney, “This Week in Hyperscaling: Deep Dive on Data Center Flexibility,” (Nov. 5, 2025), <https://elizabethkwhitney.substack.com/p/this-week-in-hyperscaling-deep-dive>.

away, every kilowatt-hour of “bridge” capacity counts. The industry is ready and willing to provide this distributed capacity as policy choices allow.

III. CONCLUSION

A wide variety of distributed capacity technologies can support the expedited interconnection of new loads without the need for physical co-location. Interconnection reform should seek to take advantage of this opportunity. The Distributed Capacity Parties thank the Commission for its consideration of these comments and its recognition of the criticality of an all-of-the-above approach that incentivizes not only new generation but also latent and quickly deployed Distributed Capacity to support the interconnection of new large loads.

Respectfully submitted,

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Dated: Nov. 21, 2025